DevOps Series
Ansible Deployment of Nginx with SSL

This is the 12th article in the DevOps series. It is a tutorial on installing Nginx with SSL. Nginx is a high performance Web server and can be used as a load balancer.

Nginx is a Web server written in C by Igor Sysoev. It can be used as a load balancer, reverse proxy and HTTP cache server. Nginx was designed to handle over 10,000 client connections and has support for TLS (transport layer security) and SSL (secure sockets layer). It requires a very low memory footprint and is IPv6-compatible. Nginx can also be used as a mail server proxy. It was first released in 2004 under a BSD-like licence.

The OpenSSL project provides a free and open source software security library that implements the SSL and TLS protocols. This library is used by applications to secure communication between machines in a computer network. The library is written in C and Assembly, and uses a dual-licence — Apache License 1.0 and a four-clause BSD licence. The library implements support for a number of ciphers and cryptographic functions. It was first released in 1998 and is widely used in Internet Web servers.

An Ubuntu 16.04.1 LTS guest virtual machine (VM) instance using KVM/QEMU is chosen to install Nginx.

```
$ cat /etc/lsb-release
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=16.04
DISTRIB_CODENAME=xenial
DISTRIB_DESCRIPTION="Ubuntu 16.04.1 LTS"
```

The default installation on the guest VM does not come with Python2, and hence you need to install this on the guest machine, manually, as shown below:

```
$ sudo apt-get update
$ sudo apt-get install python-minimal
```

The host system is a Parabola GNU/Linux-libre x86_64 system, and Ansible is installed on the host system using the distribution package manager. The version of Ansible used is 2.4.2.0 as indicated below:

```
$ ansible --version
ansible 2.4.2.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = [u'/home/shakthi/.ansible/
plugins/modules', u'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python2.7/site-
packages/ansible
  executable location = /bin/ansible
  python version = 2.7.14 (default, Sep 20 2017, 01:25:59)
[GCC 7.2.0]
```

You should add an entry to the */etc/hosts* file for the guest Ubuntu VM, as follows:

```
192.168.122.244 ubuntu
```

On the host system, let's create a project directory structure to store the Ansible playbooks, inventory and configuration files, as follows:

```
ansible/inventory/kvm/
       /playbooks/configuration/
       /playbooks/admin/
       /files/
```

An 'inventory' file is created inside the *inventory/kvm* folder that contains the following: